the information from soil sensors, such as nutrient value, drainage capacity, acidity etc, to gauge the health of the soil. It will further augment this with climatic conditions and weather predictions for that region, to suggest appropriate crops, which will give maximum yield in such a soil under such climatic conditions.

It will also take into consideration other factors like availability of seeds in that area, history of pest attacks on such crops in that region, water requirements of the crop versus water table in that region, and so on, before making the suggestions. If required, it may also suggest ways to boost soil health with appropriate nutrients.

Software platforms. Apart from individual software or apps that manage different aspects of the smart farm, we now see the emergence of a new generation of cloud-based, intelligent farm management platforms, which aim to integrate all of the farm's needs. Such platforms are not out of reach of farmers as they are often available on a software-as-a-service (SaaS) model.

Solutions galore—from soil testing to pricing

Agri-tech is often classified as pre-harvest and post-harvest technologies. Pre-harvest involves testing the soil and selecting the right crops, planting, irrigating, spraying fertilisers, monitoring the farm, mitigating pest attacks, and other farming activities.

Post-harvest technologies involve quality checking, food processing, packaging, identifying markets, logistics, pricing, and so on. The goal of technology here is to maximise profit for the farmer, while also solving the problem of food insecurity by matching demand and supply.

Hardware, software, and cloud platforms play a major role in pre- and post-harvest activities. Let us look at some interesting solutions available for Indian farmers.

Agribots. From milking cows to harvesting, threshing, and packing, robots are used for a variety of tasks on farms. Apart from global players like Universal Robots, ABB, Nexus Robotics, Agrobot Harvester, Harvest Crew, and Farmwise, there are several interesting Indian startups also working in this space.

For example, Bengaluru based TartanSense makes robots for weeding, picking, and spraying operations on farms, while Gurugram based Intello Labs makes machines that can quality-check agricultural produce, sort, grade, weigh, and pack them too. Autonomous robots from Hyderabad based XMachines can be used for spraying, weeding, cutting grass, and transplanting saplings from trays.

Drones. According to a report by BlueWeave Consulting, the Indian agriculture drones market is forecast to witness a four-fold increase by 2028, with a projected compound annual growth rate (CAGR) of more than 25% during 2022-28. Small, medium, and large drones with fixed, rotary, or hybrid wings can be used for a variety of purposes like aerial mapping, soil, and field analysis, crop monitoring, health assessment, irrigation, and spraying of nutrients and pesticides.

For example, drones equipped with normalised difference vegetation index (NDVI) imaging equipment employ detailed colour information to determine plant health. According to the report, with a team of two operators, 10 drones can plant 400,000 seeds per day.

Through the Kisan Drone Suvidha, the Indian government is promoting the use of made-in-India drones, with 5-10kg capacities, for crop assessment, spraying natural pesticides and nutrients, and carrying small loads of farm produce to the market. The Department of Agriculture & Farmers Welfare has released the standard operating procedures (SOPs), which provide concise instructions for effective and safe operations of drones for pesticide and nutrient application.

Speaking at an event last year, the Agriculture Minister Narendra Singh Tomar said that the government is promoting the use of Kisan Drones by providing a financial assistance of up to 40% (maximum ₹400,000) for the purchase of individual drones. Plus, to make drone services available to farmers on a rental basis, financial assistance of 40% (maximum ₹400,000) is provided for purchase of drones.

Certain marginalised communities and farmers in remote areas are eligible for a larger assistance. In an effort towards digitisation of land records, the government's SWAMITVA (Survey of Villages and Mapping with Improvised Technology in Village Areas) scheme also uses drones for mapping land parcels in rural inhabited areas.

The Mahindra Group, Garuda Aerospace, Thanos India, General Aeronautics, Aarav Unmanned Systems, Asteria Aerospace, BharatRohan, Dhaksha Unmanned Systems, DroneAcharya Aerial, ideaForge, and IoTechWorld are some of the main contenders in the agricultural drone space today.

Intelligence. Fasal is an IoT based, AI powered intelligence platform for horticulture crops. It captures real-time data from on-farm sensors to deliver farm-specific, crop-specific, and crop-stage-specific actionable recommendations to farmers.

Bengaluru based GramworkX has developed an IoT and AI enabled tool, which helps farmers monitor and optimise the utilisation of water. The solution uses IoT devices and a unique machine learning (ML) algorithm. The company claims that their solution provides a 20-30% yield improvement by enabling precision agriculture and consequent water savings.